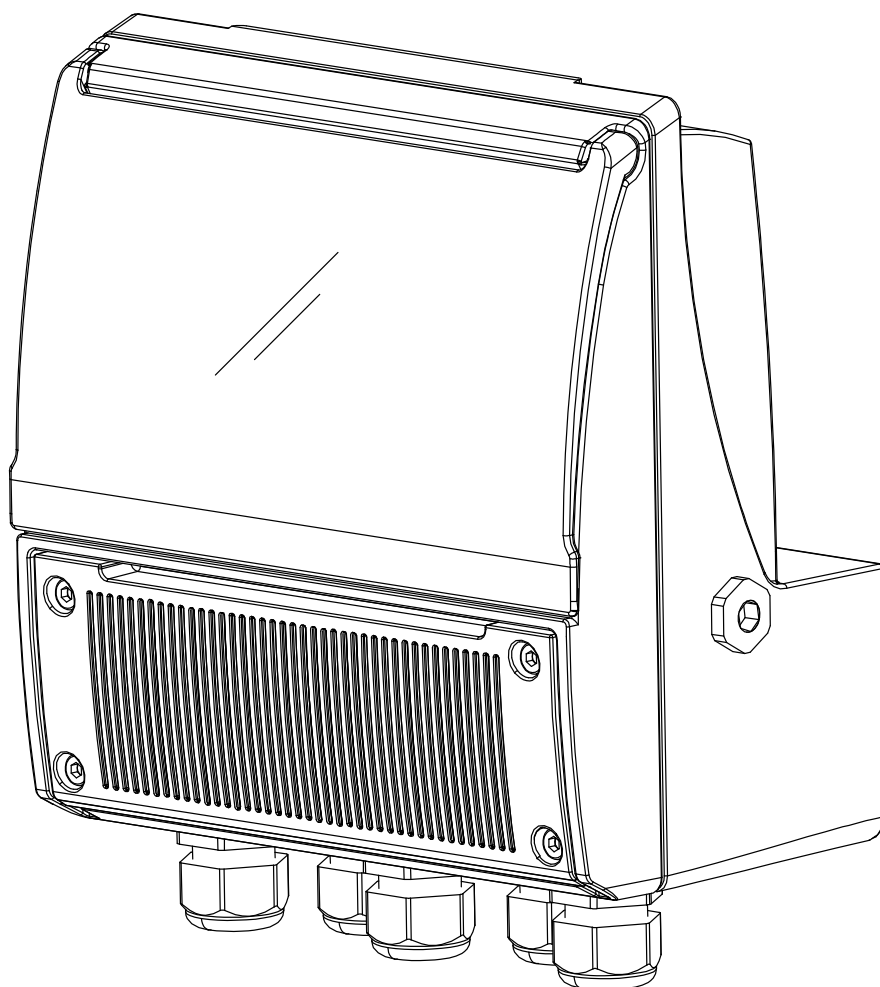




ISOMAG®

The friendly magmeter

OPERATING AND MAINTENANCE MANUAL



MV210

CE

ISOIL®
INDUSTRIA



Release number: **210_EN_IT_IS_R1_1.02.0**

The characters of file name in bold type indicate the software version which the manual refers to; it is visualized at the instrument start up, or by specific function on DIAGNOSTIC menu.

The reproduction of this manual and any supplied software is strictly forbidden.



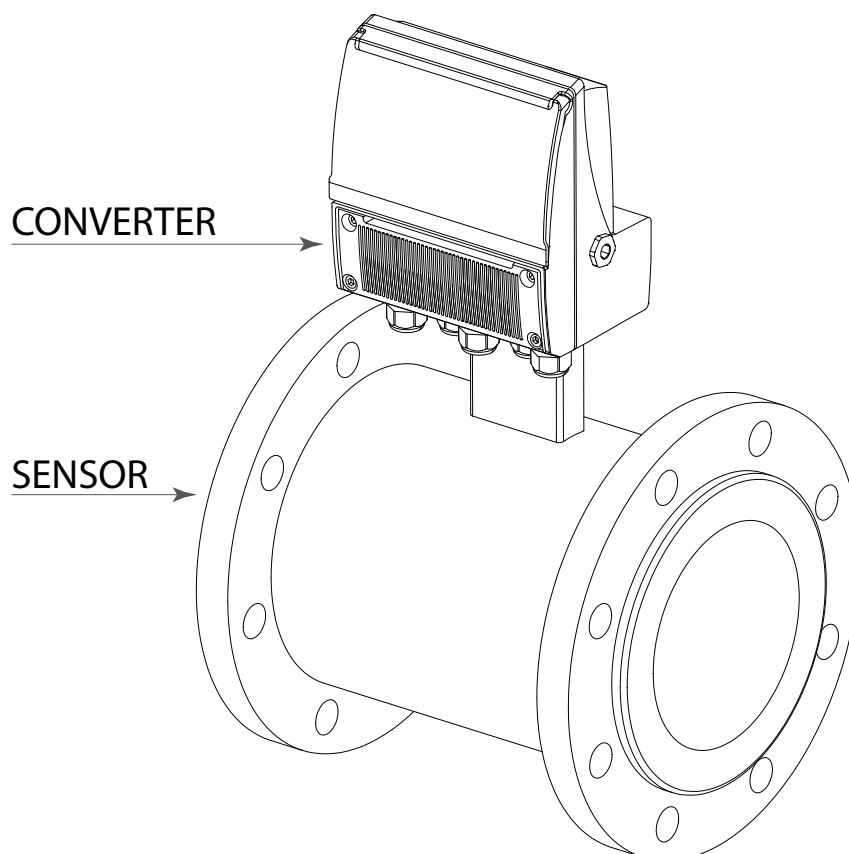
INDICE

MV210	1
SAFETY INFORMATION	3
INTRODUCTION	3
SAFETY CONVENTION	4
DATA PLATE	5
AMBIENT TEMPERATURE	5
TECHNICAL CHARACTERISTICS	5
DIMENSIONS	6
SEPARATE VERSION	7
HORIZONTAL VERSION	8
VERTICAL VERSION	8
MV210 CONSTRUCTION	10
INTERNAL LAYOUT	11
ELECTRICAL CONNECTION AND GROUNDING INSTRUCTIONS	12
INTERNAL LITHIUM BATTERY	13
DIGITAL INPUT ON/OFF OPERATION	15
OUTPUTS WIRING	16
DIGITAL OUTPUTS	16
ANALOG OUTPUTS	16
RS485 MODBUS MODULE (OPTIONAL)	17
CONNECTORS MIL	18
MEANING OF FLAGS	21
MEANING OF LED COLORS	21
ACCESS VIA KEYPAD	22
ACCESS VIA MCP INTERFACE (VIRTUAL DISPLAY)	22
ACCESS TO THE CONFIGURATION MENU	22
FLOW RATE VISUALIZATION	23
FLOW RATE ALERT	23
QUICK START MENU	23
ACCESS CODE SET : MENU 13 SYSTEM	24
RESTRICTED ACCESS SET : MENU 13 SYSTEM	24

CONVERTER ACCESS CODE	24
FUNCTIONS MENU	27
FUNCTIONS DESCRIPTION	32
MENU 1 - SENSOR	33
MENU 2 - UNITS	36
MENU 3 - SCALE	38
MENU 4 - MEASURE	40
MENU 5 - ALARMS	42
MENU 6 - INPUTS	43
MENU 7 - OUTPUTS	44
BATCHING	45
MENU 8 - COMMUNICATION	50
MV210 MODBUS PROTOCOL	50
MENU 9 - DISPLAY	53
MENU 10 - DATA LOGGER	54
MENU 11 - FUNCTION	61
MENU 12 - DIAGNOSTIC	62
MENU 13 - SYSTEM	65
MENU 14 - FILE (ONLY MCP)	71
MENU 15 - PROCESS DATA (ONLY MCP)	73
METER DATA	75
B.I.V. (BUILT-IN VERIFICATOR)	76
SOFTWARE ISOBIV	79
ALARM MESSAGES (CAUSES AND ACTIONS TO BE TAKEN)	80
ERROR CODE TEST SYSTEM OF SENSOR	81
MANUAL REVIEWS	83

INTRODUCTION

- ❑ These operating instructions and description of device functions are provided as part of the scope of supply.
- ❑ They could be modified without prior notice. The improper use, possible tampering of the instrument or parts of it and substitutions of any components not original, renders the warranty automatically void.
- ❑ The flow meter realizes a measure with liquids of conductivity greater than $5\mu\text{S}/\text{cm}$ in closed conduits, and is composed of a converter (described in this manual) and a sensor (refer to the specific manual).
- ❑ The converter could be coupled directly on the sensor (compact version) or coupled to the sensor by cable supplied with it (remote version).



SAFETY INFORMATION

Any use other than described in this manual affects the protection provided by the manufacturer and compromises the safety of people and the entire measuring system and is, therefore, not permitted. The manufacturer is not liable for damaged caused by improper or non-designated use.

- ❑ Transport the measuring device to the measuring point in the original packaging. Do not remove covers or caps until immediately before installation. In case of cartons packaging it is possible to place one above the other but no more than three cartons. In case of wooden packaging do not place one above the other.
- ❑ Disposal of this product or parts of it must be carried out according to the local public or private waste collection service regulations.
- ❑ The converter must only be installed, connected and maintained by qualified and authorized specialists (e.g. electrical technicians) in full compliance with the instructions in this Operating Instruction, the applicable norms, legal regulations and certificates (depending on the application).

- ☐ The specialists must have read and understood these Operating Instructions and must follow the instructions it contains. The Operating Instructions provide detailed information about the converter. If you are unclear on anything in these Operating Instructions, you must call the ISOIL service department.
- ☐ The converter should only be installed after have verified technical data provided in these operating instructions and on the data plate.
- ☐ Specialists must take care during installation and use personal protective equipment as provided by any related security plan or risk assessment.
- ☐ Never mount or wire the converter while it is connected to the power supply and avoid any liquid contact with the instrument's internal components. To connect remove the terminals from the terminal block.
- ☐ Before connecting the power supply check the functionality of the safety equipment.
- ☐ Repairs may only be performed if a genuine spare parts kit is available and this repair work is expressly permitted.
- ☐ For the cleaning of the device use only a damp cloth, and for the maintenance/repairs contact the service center (for details see the last page).

Before starting up the equipment please verify the following:

- ☐ Power supply voltage must correspond to that specified on the data plate
- ☐ Electric connections must be completed as described
- ☐ Ground (earth) connections must be completed as specified

Verify periodically (every 3-4 months):

- ☐ The power supply cables integrity, wiring and other connected electrical parts
- ☐ The converter housing integrity
- ☐ The suitable tightness of the sealing elements
- ☐ The front panel integrity (display and keyboard)
- ☐ The mechanical fixing of the converter to the pipe or wall stand

SAFETY CONVENTION

 DANGER ELECTRIC SHOCK	 WARNING	 PRECAUTIONS	 ATTENTION
--	--	---	--

TECHNICAL CHARACTERISTICS

ELECTRICAL CHARACTERISTIC



Converter classification: class I, IP67/68 for aluminum and PA6 housing, installation category (overvoltage) II, rated pollution degree 2.

Power supply versions	Power supply voltage	Power supply frequency	Min Power	Max power
HV	100-240V~	45-66HZ	1,5 W (Sensor only)	5 W (all Loads)
LV	24-36V---	//		
	24-36V~	45-66HZ		
LLV	12-48V---	//		

- ☐ Voltage variations must not exceed $\pm 10\%$ of the nominal one.
- ☐ Input/output insulated up to 500V.
- ☐ The output 4-20mA (optional) is electrically connected to the ON/OFF outputs and the output power supply (24V ---).
- ☐ Version LV/LLV : inrush current < 20A
Version HV : inrush current < 25A

ENVIRONMENTAL USE CONDITIONS



- ☐ The converter can be installed internally or externally
- ☐ Altitude: from -200m to 4000m (from -656 to 6560 feet)
- ☐ Humidity range: 0-100%

AMBIENT TEMPERATURE				
	Aluminium		Reinf. Nylon	
	Min*	Max	Min*	Max
°C	-20	+ 60	-10	+ 50
°F	-4	+140	+14	+122



ATTENTION

The battery will not be charged outside the below limits :

- ☐ T board MV210 < 0 °C
- ☐ T board MV210 > 50 °C

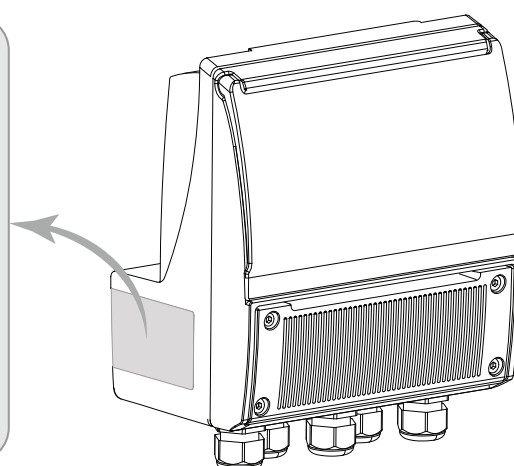
If the converter is supplied in compact version (converter over the sensor), consider the ambient temperatures more restrictive, otherwise refer to the relevant manuals.

* For discontinuous use, a thermostat heat source installation may be necessary.

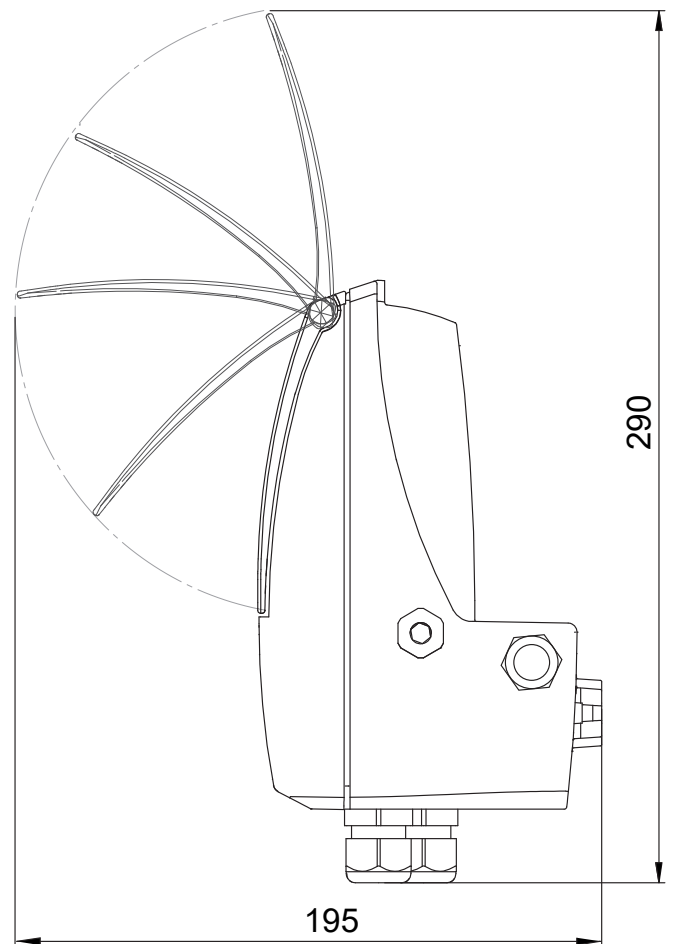
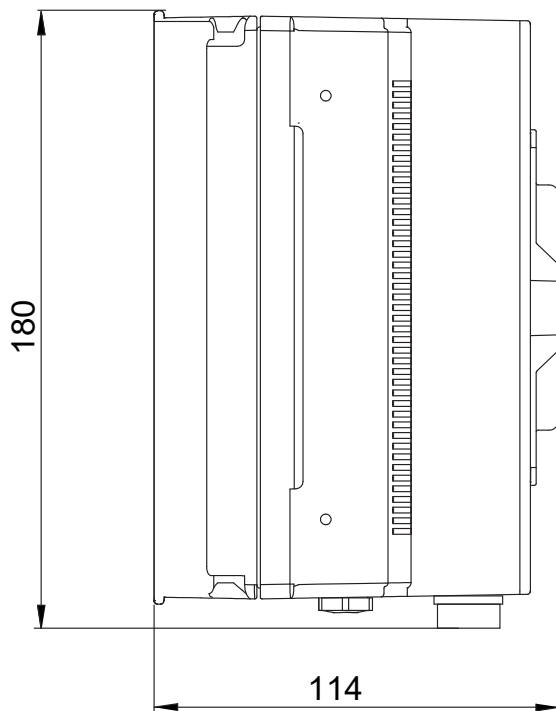
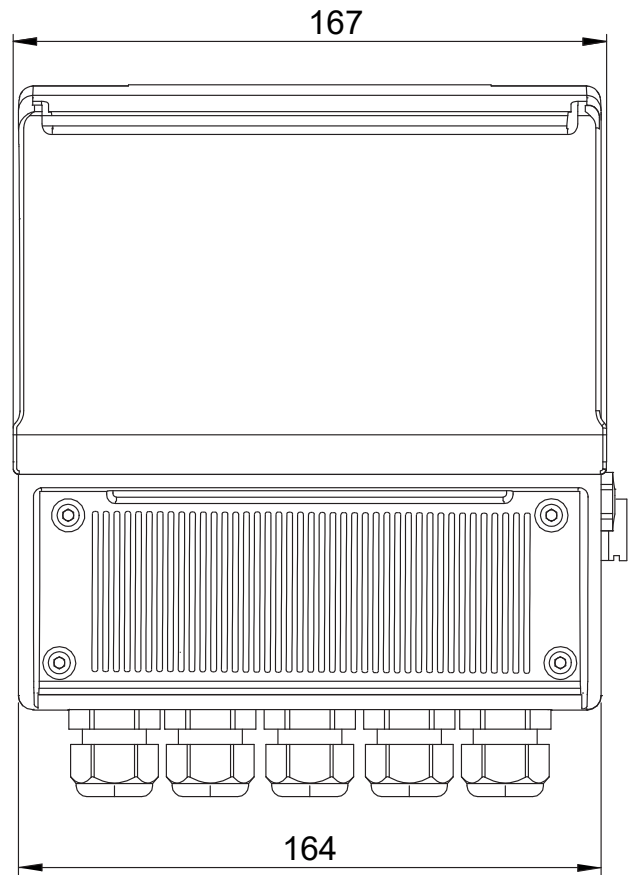
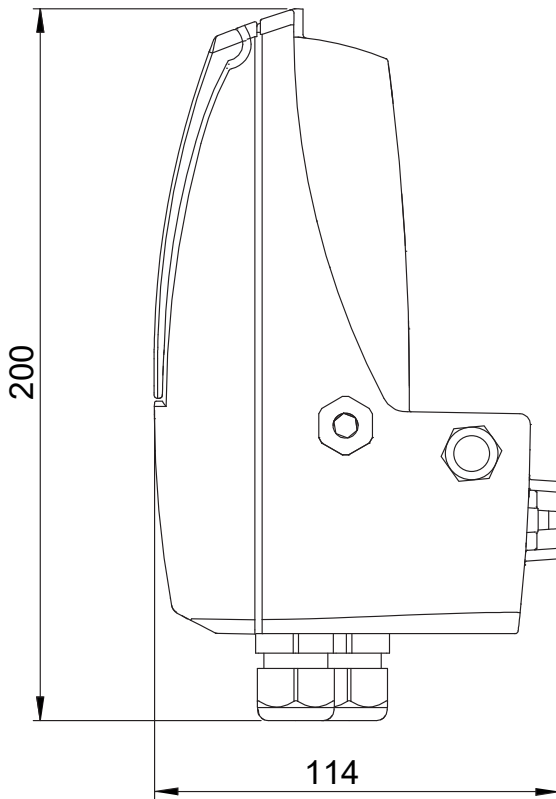
DATA PLATE

The instrument label contain the following information:

- ☐ MODEL: Convert Model
- ☐ S/N: Serial Number of the converter
- ☐ SUPPLY: Main power supply
- ☐ Hz: Supply frequency (AC)
- ☐ POWER: Maximum power consumption
- ☐ IP: Protection grade
- ☐ T: Operation temperature
- ☐ COUPLING: Serial number of sensor coupled
- ☐ ITEM: Free for user



DIMENSIONS



SEPARATE VERSION

